# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

**Full Syllabus (Chapters 1–8)** | Class 7 Maths | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 10,00,000 | Multiply and place commas. |
| 2 | **(c)** 36 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 3 | **(d)** 1 | Count the total number of decimal places. |
| 4 | **(d)** 19 | Substitute both values. |
| 5 | **(a)** $60^\circ$ | The angle and its complement add up to $90^\circ$. |
| 6 | **(c)** 1 | Digit sum must be a multiple of 3. |
| 7 | **(b)** $80^\circ$ | Share $180^\circ$ in the ratio. |
| 8 | **(b)** $3\frac{1}{2}$ | Convert both to improper fractions first. |
| 9 | **(c)** 1250000 | Divide. |
| 10 | **(c)** 27 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 11 | **(c)** 1.2 | Shift the decimal points of both numbers equally. |
| 12 | **(c)** $4x + 7y$ | Collect the like terms separately. |
| 13 | **(d)** $70^\circ$ | Find the supplement first, then halve it. |
| 14 | **(c)** 4 | Count the terms. |
| 15 | **(a)** $70^\circ$ | Exterior angle equals the sum of the two base angles. |
| 16 | **(a)** 2 | Improper form, then multiply by the reciprocal. |
| 17 | **(d)** 10000 | Divide $10^7$. |
| 18 | **(d)** 50 | Evaluate first. |
| 19 | **(a)** 11.75 | Multiply first, then add. |
| 20 | **(c)** 23 | Work out the bracket first. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Multiply and add commas (1); $35,10,000$ (1). | 1+1 |
| 22 | Innermost bracket first; Then the next bracket (1); Then the rest in order; Value $= 18$ (1). | 1+1 |
| 23 | $5.6 + 3.45 = 9.05$ (1); $9.05 - 2.8 = 6.25$ (1). | 1+1 |
| 24 | Expand the bracket: $3x + 12$ (1); Add $5x$: $8x + 12$ (1). | 1+1 |
| 25 | Let the complement be $y$; the angle is $y + 20$ and $y + (y + 20) = 90$; $2y = 70$, so $y = 35$ (1); The angle is $55^\circ$ (1). | 1+1 |
| 26 | Digit sum $= 10 + *$; Must be a multiple of 9, so $* = 8$ (1); Number $= 5814$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | In 8 years: $1,25,000 \times 8 = 10,00,000$ km (1); Trains needed $= 60,00,000 \div 10,00,000 = 6$ (2). | 1+2 |
| 28 | Cost $= 50 \times 40 = 2000$; Sales $= 30 \times 60 + (50 - 30) \times 35 = 2500$ (1); Result $= 500$ (1); Profit (1). | 1+1+1 |
| 29 | Distance $= 12.5 \times 8 = 100$ km (1); Litres $= 75 \div 12.5 = 6$ (2). | 1+2 |
| 30 | Sum $= 10n + 3$ (1); Difference $= 4n + 7$ (1); At $n = 4$: sum $= 43$ and difference $= 23$ (1). | 1+1+1 |
| 31 | Adjacent angles form a linear pair: 5 parts $= 180^\circ$ (1); One part $= 36^\circ$, so the adjacent angles are $36^\circ$ and $144^\circ$ (1); Vertically opposite angles are equal, so the four angles are $36^\circ$, $144^\circ$, $36^\circ$, $144^\circ$ (1). | 1+1+1 |
| | **OR:** $7621 - 1267 = 6354$; $6543 - 3456 = 3087$; $8730 - 0378 = 8352$; $8532 - 2358 = 6174$ (1); Steps needed $= 4$ (2). | 1+2 |
| 32 | One part $= 180^\circ \div 9 = 20^\circ$; Angles $40^\circ$, $60^\circ$, $80^\circ$ (1); Exterior angle $= 180^\circ - 80^\circ$ (1); $= 100^\circ$ (1). | 1+1+1 |
| | **OR:** Area $= \frac{3}{2} \times \frac{9}{4}$; $= 3\frac{3}{8}$ m$^2$ (1); Perimeter $= 2(1\frac{1}{2} + 2\frac{1}{4})$; $= 2 \times 3\frac{3}{4}$ (1); $= 7\frac{1}{2}$ m (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | 2nd year $= 14,00,000$ (1); 3rd year $= 15,50,000$ (1); 4th year $= 17,00,000$ (1); Total $= 59,00,000$ (2). | 1+1+1+2 |
| **34** | Chairs after discount $= 40 \times (450 - 50) = 16000$ (1); Tables $= 10 \times 1200 = 12000$ (1); Total $= 28000$ (1); Expression: $40 \times (450 - 50) + 10 \times 1200$ (2). | 1+1+1+2 |
| **35** | $9.5 \div 1.2$ gives 7 whole pieces (1); Length used $= 7 \times 1.2 = 8.4$ m (2); Left over $= 9.5 - 8.4 = 1.1$ m (2). | 1+2+2 |
| **36** | Sales $= 48x + 36y$ (1); At the given values: $480 + 540$ (2); Total sales $=$ ₹$1020$ (2). | 1+2+2 |
| | **OR:** They are supplementary: $5$ parts $= 180^\circ$ (1); One part $= 36^\circ$, so the angles are $72^\circ$ and $108^\circ$ (2); Linear pair with the larger: $180^\circ - 108^\circ = 72^\circ$ (2). | 1+2+2 |
| **37** | Sequence: $7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $= 52$ (1); Even terms $= 11$ (1); Odd terms $= 6$ (1); Total terms $= 17$ (1). | 1+1+1+1+1 |
| | **OR:** Equal sides: $2x + 1 = 3x - 4$, so $x = 5$ (1); Each equal side $= 11$ cm (1); Base $= 5 + 3 = 8$ cm (1); Perimeter $= 11 + 11 + 8 = 30$ cm (1); Existence: $11 + 8 = 19 > 11$ and $11 + 11 = 22 > 8$ (1). | 1+1+1+1+1 |
| **38** | Day 1 $= \frac{1}{4} \times 240 = 60$ m$^2$ (1); Remainder $= 180$ m$^2$ (1); Day 2 $= \frac{2}{3} \times 180 = 120$ m$^2$ (1); Day 3 $= 180 - 120$ (1); $= 60$ m$^2$ (1). | 1+1+1+1+1 |
| | **OR:** Full days: $12,50,000 \div 40,000$ gives 31 full days (1); Remaining $= 10,000$ m (2); Total days to finish: 32 (2). | 1+2+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

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## ANSWER KEY & MARKING SCHEME SET 2

**Full Syllabus (Chapters 1–8)** | Class 7 Maths | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $60^\circ$ | They add up to $180^\circ$. |
| 2 | **(c)** 6174 | Largest arrangement minus smallest arrangement. |
| 3 | **(d)** $75^\circ$ | Exterior angle $= \angle A + \angle C$. |
| 4 | **(a)** $1$ | Work from left to right. |
| 5 | **(b)** 19,99,999 | 999999 + 1000000. |
| 6 | **(c)** 9 | Divide the product by the multiplier, then subtract. |
| 7 | **(a)** 1.25 | Divide the total by the number of persons. |
| 8 | **(a)** $5x + 7$ | Subtract the $x$-terms and the constants separately. |
| 9 | **(c)** $90^\circ$ | The angles around a point add up to $360^\circ$. |
| 10 | **(b)** 9 | The first half decides the number. |
| 11 | **(c)** 7 | Count whole numbers strictly between the difference and the sum. |
| 12 | **(b)** $21$ | Number = part ÷ fraction. |
| 13 | **(a)** 1800000 | Round first. |
| 14 | **(c)** 325 | Total minus reserved. |
| 15 | **(b)** 12 | Multiply first, then subtract. |
| 16 | **(b)** 43 | Substitute the value of $n$. |
| 17 | **(d)** $75^\circ$ | Let the angle be $y$; its supplement is $y + a$. |
| 18 | **(d)** 60 | Use the middle number times the count when the count is odd. |
| 19 | **(c)** $70^\circ$ | Substitute $d$. |
| 20 | **(a)** $2\frac{1}{2}$ | Time = $1 \div$ rate. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Case 1: $6$ is an equal side, base $= 26 - 2 \times 6 = 14$; not valid, since $6 + 6 = 12$ is not greater than $14$; Case 2: $6$ is the base, equal sides $= (26 - 6) \div 2 = 10$; valid (1); Sides (cm): $10, 10, 6$ (1). | 1+1 |
| 22 | Left to right: $\frac{2}{3} \times \frac{9}{10} = \frac{3}{5}$; Then divide: $\frac{3}{5} \times \frac{5}{3}$ (1); Answer $1$ (1). | 1+1 |
| 23 | Divide step by step (1); $36,00,000$ (1). | 1+1 |
| 24 | $6 \times (9 + 8)$ (1); Value $= 102$ (1). | 1+1 |
| 25 | $1.2 \times 0.5 = 0.6$ (1); $0.6 \div 0.04 = 15$ (1). | 1+1 |
| 26 | $(9x + 2) - (3x - 5)$ (1); Answer: $6x + 7$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | (i) $34,57,000$ (1); (ii) $35,00,000$ (1); Difference $= 43,000$ (1). | 1+1+1 |
| 28 | Brackets first; Multiplication (1); Final answer (1); Value $= 180$ (1). | 1+1+1 |
| 29 | Cost $= 8 \times 7.5 = 60$ (1); Balance $= 100 - 60 = 40$ (2). | 1+2 |
| 30 | For $9$ triangles: $2(9) + 1 = 19$ (1); For $14$ triangles: $2(14) + 1 = 29$ (2). | 1+2 |
| 31 | Let the complement be $y$; the angle is $2y + 6$ and $y + (2y + 6) = 90$ (1); $3y = 84$, so $y = 28$ (1); The angle is $62^\circ$ and its complement is $28^\circ$ (1). | 1+1+1 |
| | **OR:** $11(a + b) = 99$, so $a + b = 9$; $a - b = 3$ (1); $a = 6$, $b = 3$ (1); Number $= 63$ (1). | 1+1+1 |
| 32 | Equal sides, so $\angle Q = \angle R$; $3x - 10 = x + 6$ so $x = 8$ (1); $\angle Q = \angle R = 14^\circ$; $\angle P = 180^\circ - 28^\circ = 152^\circ$ (1); Exterior angle at P $= 28^\circ$ (1). | 1+1+1 |
| | **OR:** Spent $= \frac{1}{4} \times 240 = 60$; Left $= 240 - 60 = 180$ (1); Each $= 180 \div 6$ (1); $=$ ₹30 (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | Evaluate brackets first (1); Then multiplication/division (1); Value $= 465$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |
| **34** | Time $= 90 \div 7.5 = 12$ minutes (1); After 6 minutes: $7.5 \times 6 = 45$ litres (2); More needed $= 90 - 45 = 45$ litres (2). | 1+2+2 |
| **35** | Mother now $= 3r + 4$ (1); After 5 years: Raju $= r + 5$, mother $= 3r + 9$ (1); Sum $= 4r + 14$ (1); At $r = 10$: 54 years (2). | 1+1+1+2 |
| **36** | Total parts $= 10$ and $360^\circ \div 10 = 36^\circ$ (1); The angles are $36^\circ$, $72^\circ$, $108^\circ$, $144^\circ$ (2); Kinds: $36^\circ$ acute, $72^\circ$ acute, $108^\circ$ obtuse, $144^\circ$ obtuse (2). | 1+2+2 |
| | **OR:** Palindrome form $1b1$ with digit sum $= 2 \times 1 + b$ (1); Need $2 + b$ to be a multiple of 3 (1); Possible $b$: $1, 4, 7$ (1); Palindromes: $111, 141, 171$ (2). | 1+1+1+2 |
| **37** | $2, 4, 6$: 2+4=6 not > 6, no triangle; $2, 4, 7$: 2+4=6 not > 7, no triangle; $2, 6, 7$: 2+6=8 > 7, triangle; $4, 6, 7$: 4+6=10 > 7, triangle (2); Number of triangles $= 2$ (3). | 2+3 |
| | **OR:** Syrup $= \frac{2}{5} \times \frac{5}{2} = 1$ L (1); Water $= 2\frac{1}{2} - 1 = 1\frac{1}{2}$ L (1); Glasses $= 2\frac{1}{2} \div \frac{1}{4}$ (1); $= 10$ (1); Full glasses: 10 (1). | 1+1+1+1+1 |
| **38** | 2nd year $= 26,50,000$ (1); 3rd year $= 29,00,000$ (1); 4th year $= 31,50,000$ (1); Total $= 1,11,00,000$ (2). | 1+1+1+2 |
| | **OR:** Chairs after discount $= 30 \times (600 - 100) = 15000$ (1); Tables $= 8 \times 1500 = 12000$ (1); Total $= 27000$ (1); Expression: $30 \times (600 - 100) + 8 \times 1500$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

**Full Syllabus (Chapters 1–8)** | Class 7 Maths | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 86400 | 1 hour = 3600 seconds. |
| 2 | **(b)** $20 - 5 \times 2$ | Evaluate each option. |
| 3 | **(b)** 3.45 | 1 m = 100 cm, so write the smaller unit as a decimal part. |
| 4 | **(d)** $3x + 5y$ | Multiply each price by its quantity. |
| 5 | **(c)** parallel | Check the angle relation. |
| 6 | **(a)** even | Test with small values of $n$. |
| 7 | **(d)** SAS | Check which parts are given and where they sit. |
| 8 | **(c)** $\frac{3}{4} \div \frac{1}{8}$ | Compare by computing. |
| 9 | **(a)** 3650 | Multiply. |
| 10 | **(a)** 15 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 11 | **(b)** 3.46 | Look at the thousandths digit. |
| 12 | **(a)** 15 | $a^2 = a \times a$. |
| 13 | **(d)** $100^\circ$ | Larger + smaller = $180^\circ$ and larger − smaller = $d$. |
| 14 | **(d)** 3 but not 9 | Use the digit-sum tests. |
| 15 | **(c)** 24 | Add the three sides. |
| 16 | **(a)** 105 | Distance = rate × quantity. |
| 17 | **(c)** 34,50,000 | Multiply and place commas. |
| 18 | **(a)** 28 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 19 | **(a)** 0.37 | A number between them is greater than the smaller and less than the larger. |
| 20 | **(d)** 11 | Work out the numerator, then divide. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Larger + smaller = 180 and larger − smaller = 20; Larger = $(180 + 20) \div 2 = 100$ (1); The angles are $100^\circ$ and $80^\circ$ (1). | 1+1 |
| 22 | Middle odd number = $51 \div 3 = 17$; Others differ by 2: 15 and 19 (1); Numbers: 15, 17, 19; Check sum = 51 (1). | 1+1 |
| 23 | Exterior = sum of the interior opposite angles; $4x - 10 = 3x + 30$ so $x = 40$ (1); Exterior angle = $150^\circ$ (1). | 1+1 |
| 24 | Number = $27 \div \frac{3}{4}$; = 36 (1); $\frac{1}{4}$ of it = 9 (1). | 1+1 |
| 25 | 1 day = 24 × 60 = 1440 minutes (1); $30 \times 1440 = 43,200$ (1). | 1+1 |
| 26 | First = 60; Second = 60 (1); Equal (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | Exact = 45,67,900 (1); Estimate = 46,00,000 (1); The estimate is close to the exact answer (1). | 1+1+1 |
| 28 | Try bracketing different parts (1); Correct: $(2 + 3) \times 4 + 5$ (1); Check value 25 (1). | 1+1+1 |
| 29 | Compare whole parts, then tenths, then hundredths (1); Descending order: 4.55, 4.5, 4.05, 4.005 (1); Difference = $4.55 - 4.005 = 0.545$ (1). | 1+1+1 |
| 30 | Fare = $45 + 11k$ (1); For 8 km: $45 + 11(8) = 133$ (1); For 13 km: $45 + 11(13) = 188$ (1). | 1+1+1 |
| 31 | 4 parts = $180^\circ$, one part = $45^\circ$ (1); The angles are $45^\circ$ and $135^\circ$ (1); Complement of the smaller angle = $90^\circ - 45^\circ = 45^\circ$ (1). | 1+1+1 |
| | **OR:** Middle number = $75 \div 5 = 15$; First number = $15 - 2 = 13$ (1); Numbers: 13, 14, 15, 16, 17 (1); Check sum = 75 (1). | 1+1+1 |
| 32 | Corresponding sides of congruent triangles are equal, so $AB = DE$; $5x - 2x = 18 - 3$, so $3x = 15$ and $x = 5$ (1); $DE = AB = 28$ cm; Perimeter = $AB + BC + AC = 28 + 7 + 9$ (1); = 44 cm (1). | 1+1+1 |

| | | |
|---|---|---|
| | **OR:** Bracket: $\frac{1}{2}+\frac{1}{3}=\frac{5}{6}$ (1); Multiply: $\frac{5}{6}\times\frac{6}{5}$ (1); $=1$ (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | $7.8\div0.9$ gives $8$ whole pieces (1); Length used $=8\times0.9=7.2$ m (2); Left over $=7.8-7.2=0.6$ m (2). | 1+2+2 |
| **34** | Sales $=45x+30y$ (1); At the given values: $900+300$ (2); Total sales $=₹1200$ (2). | 1+2+2 |
| **35** | They are supplementary: $6$ parts $=180^\circ$ (1); One part $=30^\circ$, so the angles are $30^\circ$ and $150^\circ$ (2); Linear pair with the larger: $180^\circ-150^\circ=30^\circ$ (2). | 1+2+2 |
| **36** | Sequence: $20, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $=20$ (1); Even terms $=6$ (1); Odd terms $=2$ (1); Total terms $=8$ (1). | 1+1+1+1+1 |
| | **OR:** Equal sides: $2x+2=3x-3$, so $x=5$ (1); Each equal side $=12$ cm (1); Base $=5+6=11$ cm (1); Perimeter $=12+12+11=35$ cm (1); Existence: $12+11=23>12$ and $12+12=24>11$ (1). | 1+1+1+1+1 |
| **37** | Day 1 $=\frac{2}{5}\times180=72$ m$^2$ (1); Remainder $=108$ m$^2$ (1); Day 2 $=\frac{1}{3}\times108=36$ m$^2$ (1); Day 3 $=108-36$ (1); $=72$ m$^2$ (1). | 1+1+1+1+1 |
| | **OR:** Full days: $24,50,000\div65,000$ gives 37 full days (1); Remaining $=45,000$ m (2); Total days to finish: 38 (2). | 1+2+2 |
| **38** | Evaluate brackets first (1); Then multiplication/division (1); Value $=80$ (1); A suitable story is consistent with the operations (2). | 1+1+1+2 |
| | **OR:** Time $=72\div4.8=15$ minutes (1); After $9$ minutes: $4.8\times9=43.2$ litres (2); More needed $=72-43.2=28.8$ litres (2). | 1+2+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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